

Have Open Borders Contributed to European Productivity Growth?

An Economic Impact Assessment of the Schengen Agreement 30 Years Later

Achille Marteret & Einar Fæste

ECO433: Empirical Strategies for Causal Analysis

Research Question

Core Question

Did the Schengen Agreement increase labor productivity across Western Europe?

- Schengen abolished border controls - enhancing labor and goods mobility.
- Hypothesis: **Free movement $\Rightarrow$ higher productivity through specialization and integration.**

Why It Matters

- 3.5 million people cross a Schengen border daily (EC, 2025).
- Productivity growth in Europe has stagnated compared to the U.S. and China.
- Border reintroductions threaten competitiveness.

Our Contribution

Causal estimate of Schengen's productivity impact using modern econometric tools.

Data and Variables

- **Data source:** Penn World Tables (PWT 10.0)
- **Sample:** Western European countries (1985–2007)
- **Outcome variables:**

$$Y_{it}(h) = \log(\text{GDP per hour})$$

$$Y_{it}(w) = \log(\text{GDP per worker})$$

- **Treatment:** Entry into Schengen (staggered by country)
- Exclude Eastern Europe to avoid catch-up bias.

Empirical Design: Staggered Difference-in-Differences

- Following Callaway & Sant'Anna (2021)
- Exploit staggered rollout of Schengen (1995–2001)
- Control group: countries not yet treated (“NYT”)

Model:

$$ATT(g, t) = E[Y_{it}(g) - Y_{it}(\infty) | G_i = g], \quad t \geq g$$

Average across cohorts:

$$ATT(e) = \sum_g w_{g,e} ATT(g, g + e)$$

Key assumptions: No anticipation effects; Parallel pre-trends.

Bootstrap Method for Inference

- To address the low number of clusters in our panel (15 countries), we apply a **multiplier bootstrap** at the country level.
- This method resamples residuals while preserving within-country dependence.
- It improves the **finite-sample coverage** of standard errors and reduces bias in the estimated p -values.
- In our case, it lowers the p -value for $\log(GDP_{per\ worker})$ from 2.5% to 2.1%.

Anticipation Effects and Identification

• Our Approach:

- We use each country's *signing dates* into Schengen as the treatment date.
- We test for pre-trends: no significant increase in productivity is observed before entry.
- Figure evidence shows effects emerging only **around the implementation year**, not earlier.

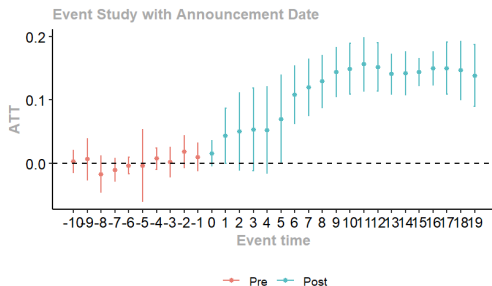


Figure: No visible anticipatory jump prior to Schengen entry

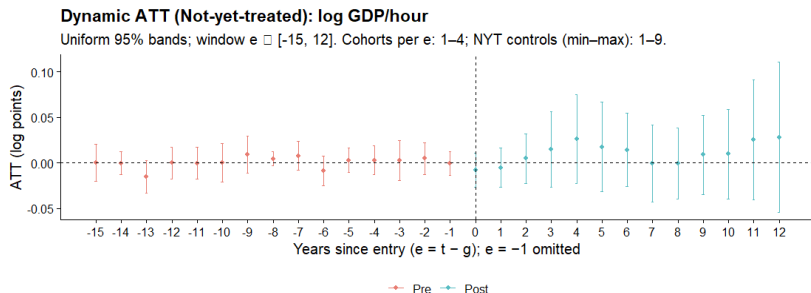
Main Results Overview

- **GDP per hour:** No significant impact.
- **GDP per worker:** Significant positive effect after 5 years.
- Suggests an adjustment period post-border removal.

Summary

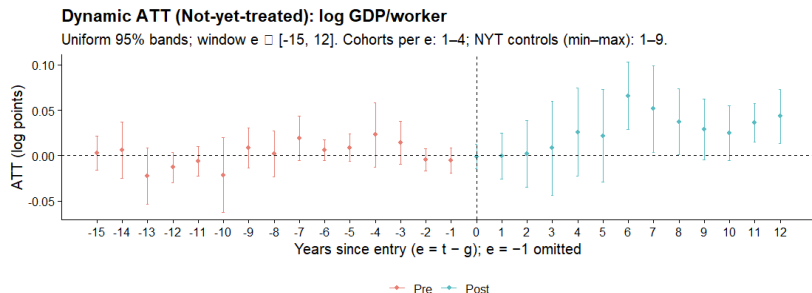
Productivity gains emerged gradually - workers and firms needed time to adapt to borderless labor markets and supply chains.

Visual Evidence: GDP per Hour



Effect of Schengen on log GDP per hour (PPP adjusted).

Visual Evidence: GDP per Worker



Effect of Schengen on log GDP per worker (PPP adjusted). Significant increase visible from year 5 onward.

Estimated Treatment Effects

Overall ATT, Callaway & Sant'Anna (NYT controls)

outcome	ATT (log)	SE	95% CI	ATT (%)	p-value
log GDP/hour	0.010	(0.011)	[-0.012, 0.032]	1.0	0.382
log GDP/worker	0.024*	(0.010)	[0.004, 0.044]	2.4	0.021

- $\Rightarrow$ 2.4% productivity gain per worker post-Schengen.
- Statistically significant at 5% level.

Interpretation and Limitations

- Supports the view that openness boosts productivity.
- Gains emerge gradually - reflecting adaptation in labor markets and firms.
- **Limitations:**
 - Excludes Eastern Europe (limited external validity).
 - Possible spillover effects across neighbors.
 - Period ends in 2007 (avoids 2008 crisis confounding).

Policy Implications

- Reintroducing border controls likely reduces productivity and competitiveness.
- Schengen remains a cornerstone for EU economic integration.
- Productivity benefits justify maintaining open borders.

Conclusion

- Schengen led to a **2.4% rise in GDP per worker** across Western Europe (1995–2007).
- Border removal fostered labor mobility, specialization, and integration.
- Evidence supports the idea that openness drives long-term productivity.

Appendix

Variable	Effect (log points)	p-value
GDP per hour	+0.010	0.394
GDP per worker	+0.024	0.025**

- $\Rightarrow$ 2.4% productivity gain per worker post-Schengen.
- Statistically significant at 5% level.

References



Baker, A., Callaway, B., Cunningham, S., Goodman-Bacon, A., & Sant'Anna, P. H. C. (2025). *Difference-in-Differences Designs: A Practitioner's Guide*. <https://doi.org/10.48550/arxiv.2503.13323>



Callaway, B., & Sant'Anna, P. H. C. (2021). *Difference-in-Differences with multiple time periods*. *Journal of Econometrics*, 225(2), 200–230. <https://doi.org/10.1016/j.jeconom.2020.12.001>



Davis, D., & Gift, T. (2014). *The Positive Effects of the Schengen Agreement on European Trade*. *World Economy*, 37(11), 1541–1557. <https://doi.org/10.1111/twec.12158>



Draghi, M. (2024). *The Future of European Competitiveness: A Competitiveness Strategy for Europe*. Report to the European Commission. Brussels: European Commission. https://commission.europa.eu/topics/eu-competitiveness/draghi-report_en



European Commission. (2025a). *Schengen Area*. https://home-affairs.ec.europa.eu/policies/schengen/schengen-area_en



European Commission. (2025b). *History of Schengen*. https://home-affairs.ec.europa.eu/policies/schengen/schengen-area/history-schengen_en



Feenstra, R. C., Inklaar, R., & Timmer, M. P. (2015). *The Next Generation of the Penn World Table*. *American Economic Review*, 105(10), 3150–3182. <https://www.rug.nl/ggdc/productivity/pwt/related-research>



Gustafsson, P., & Segerstrom, P. (2010). *Trade Liberalization and Productivity Growth*. *Review of International Economics*, 18(2), 207–228. <https://doi.org/10.1111/j.1467-9396.2010.00878.x>



Melitz, M. J., & Redding, S. J. (2014). *Missing Gains from Trade?* *American Economic Review*, 104(5), 317–321. <https://doi.org/10.1257/aer.104.5.317>



Orvedal, L. (2025). *Lecture 3: International Trade – Theory [Lecture Slides]*. NHH, Canvas. https://nhh.instructure.com/courses/2717/files/455034?module_item_id=153331